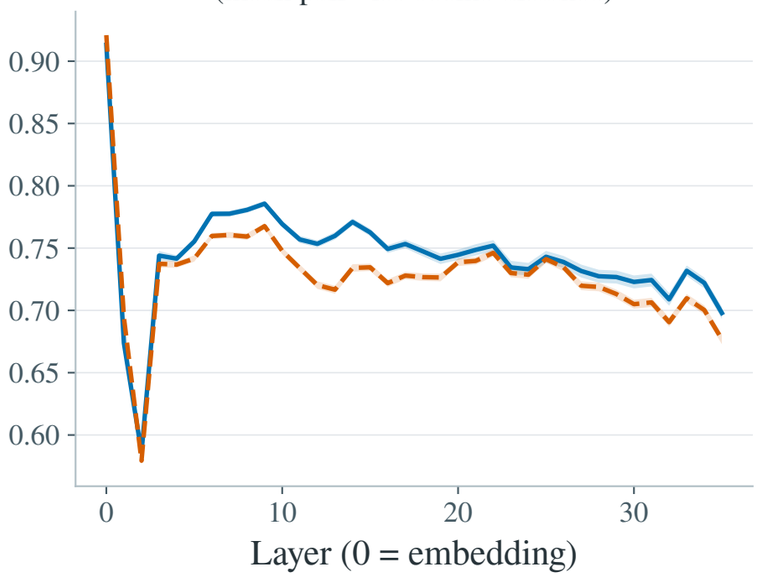
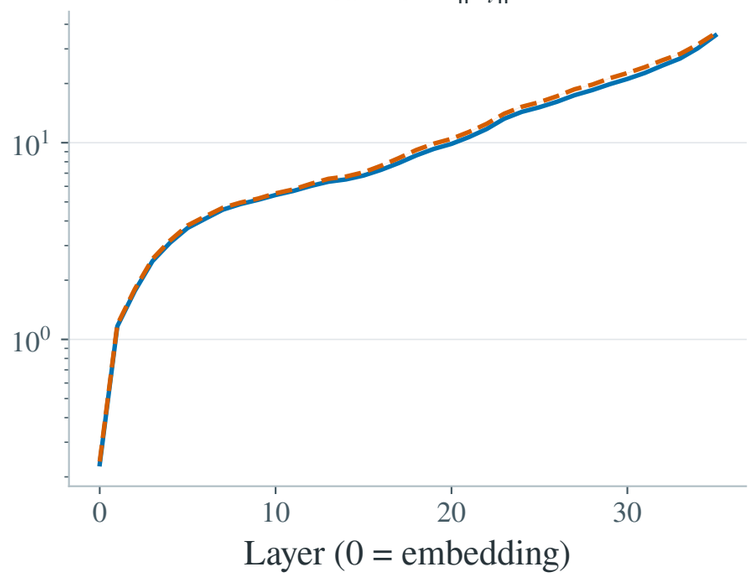
Ministral-8B, drlXDomainLen_wiki_500_tokens: layer profiles

Human — Machine

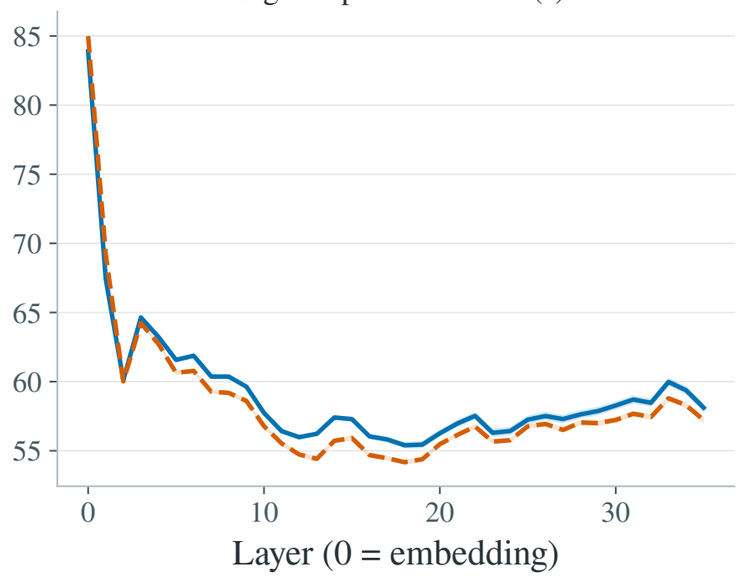
Representation dispersion (mean pairwise cosine distance)



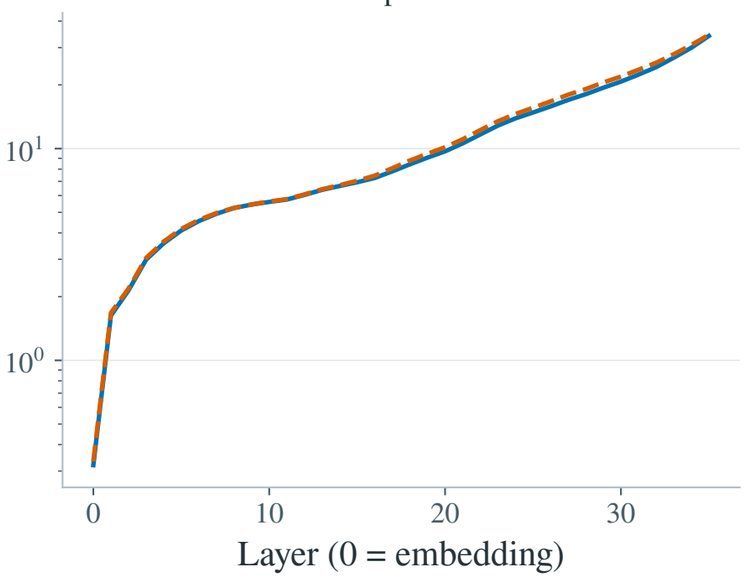
Mean norm $\|h_t\|$



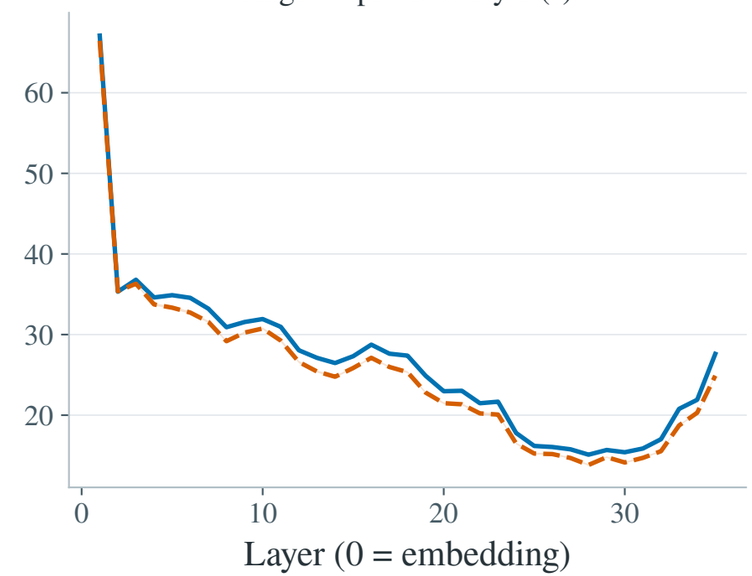
Angle to previous token (°)



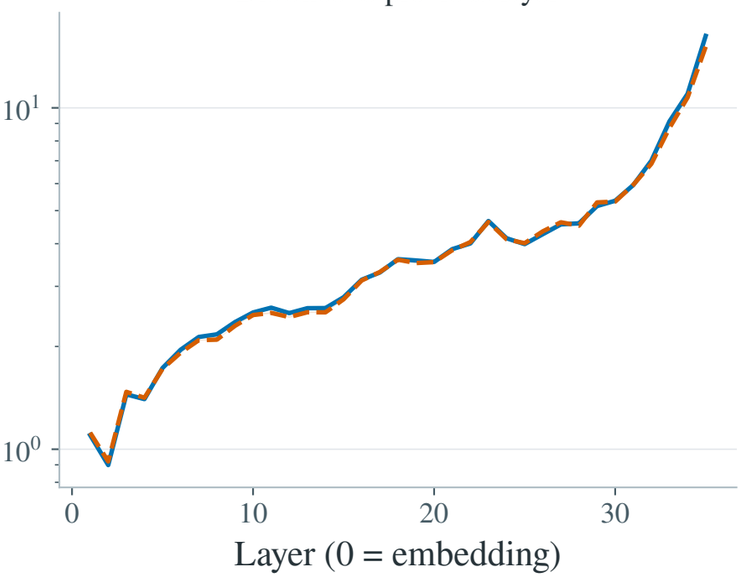
Distance to previous token



Angle to previous layer (°)



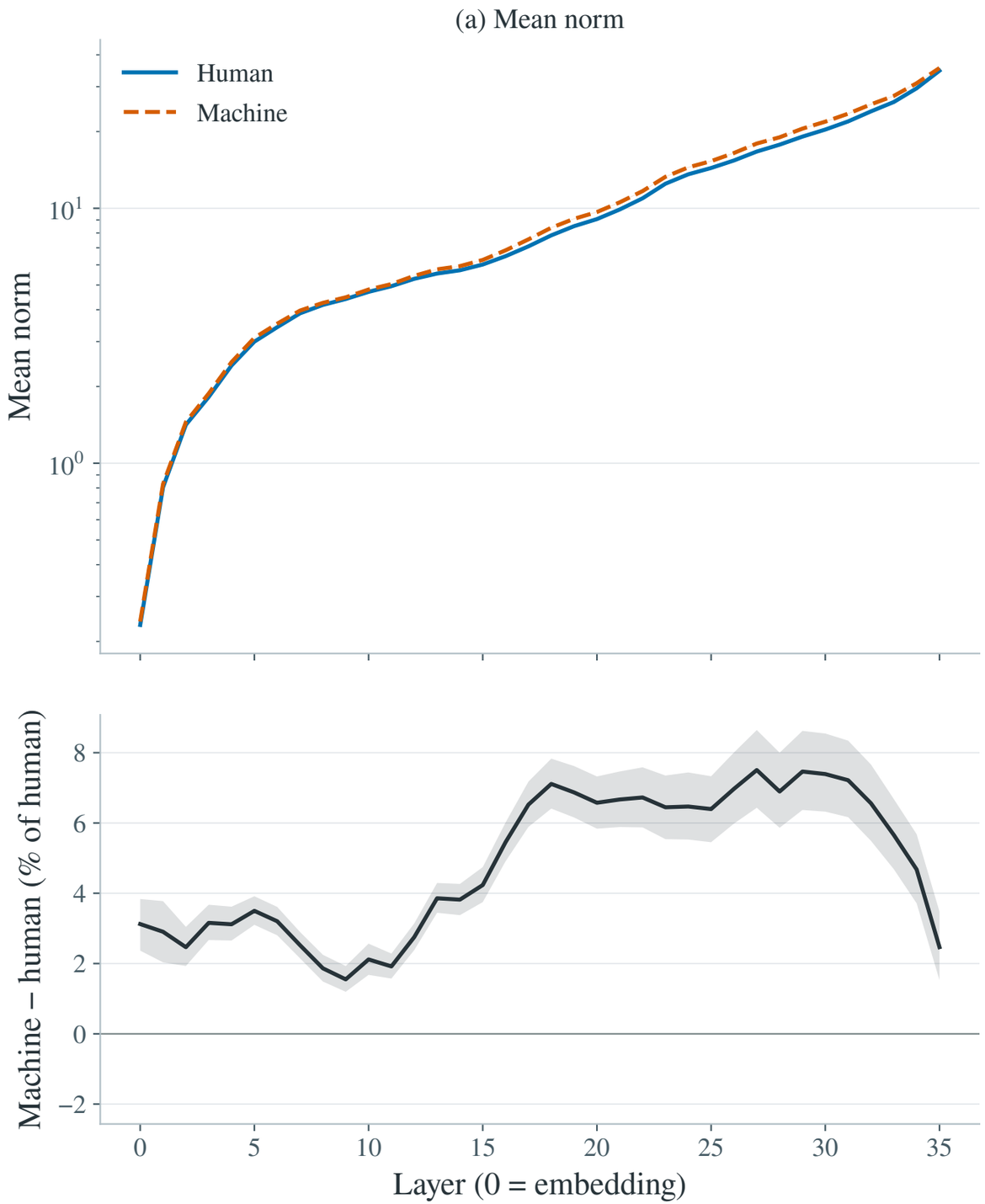
Distance to previous layer

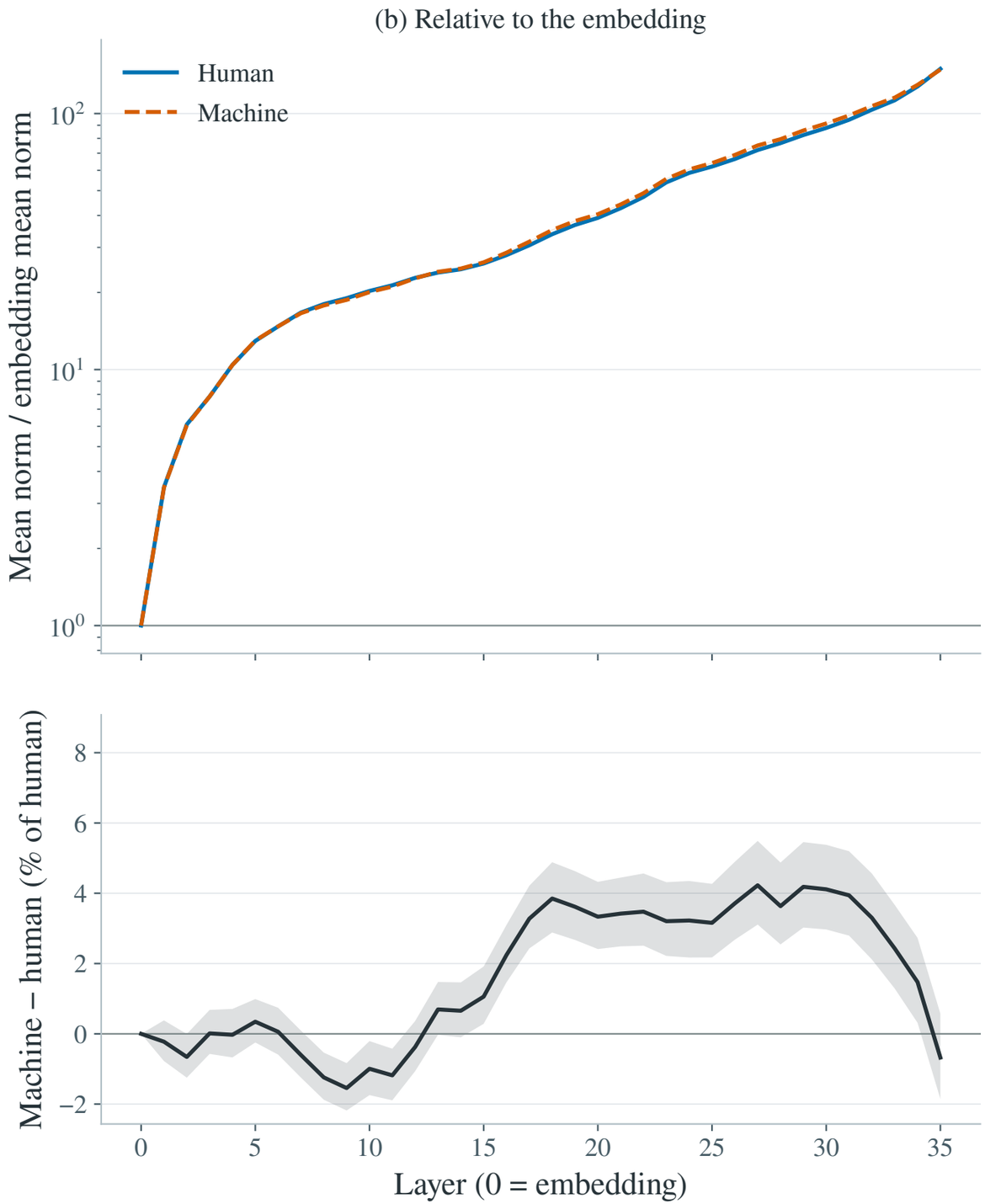


Mean over texts per layer, band = 95% CI of the mean. Log y for norm and distances. Final-norm layer left out.

Ministral-8B, drlXDomainLen_wiki_500_tokens: norm growth and the machine − human gap

Norm without outlier tokens: per text and layer, tokens with norm > 2 × the text's median token norm are excluded (on average 2.4 tokens per text, 0.46%, layers 1+)

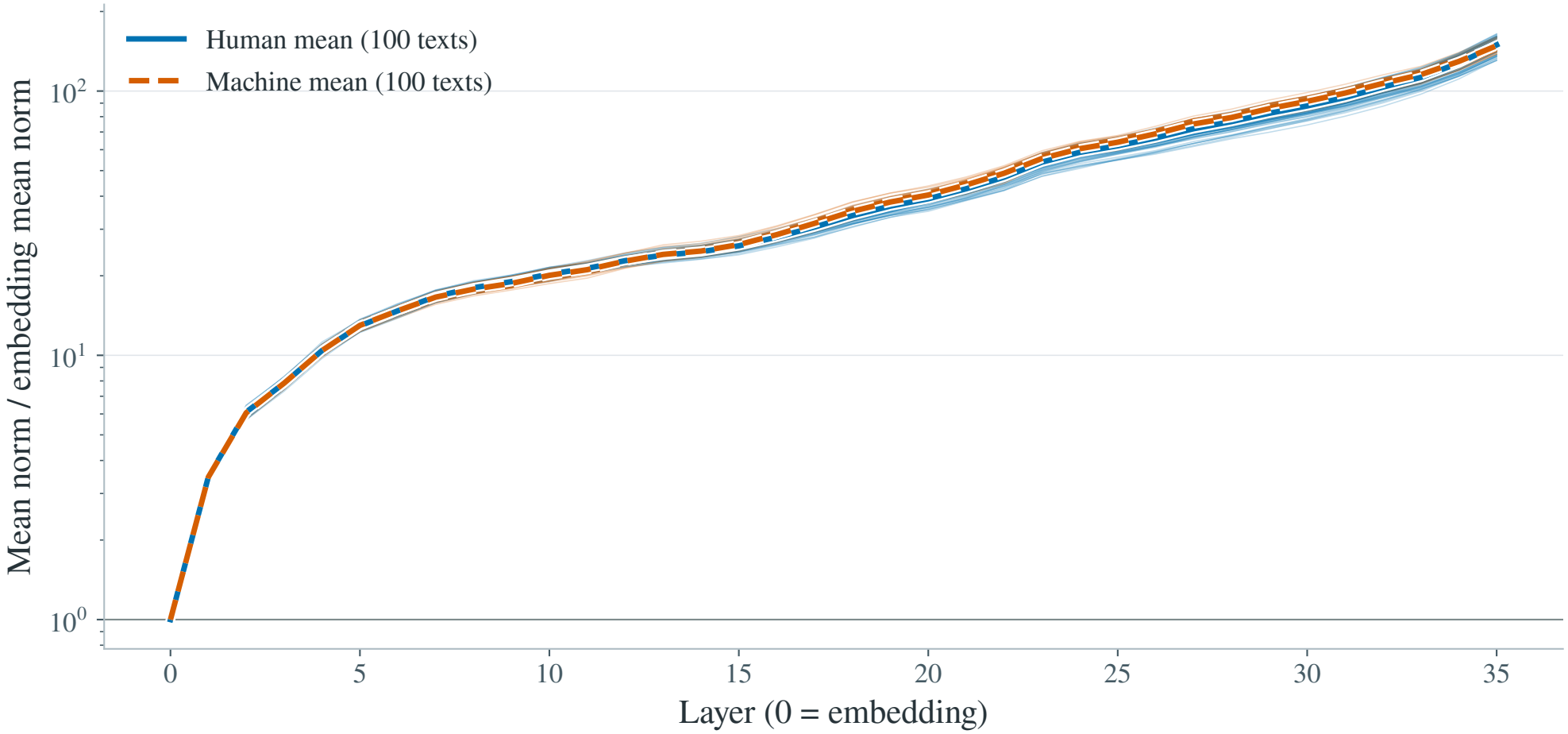




Top: mean over texts, band = 95% CI, log y; (b) divides each text by its own embedding-layer norm. Bottom: gap in % of the human mean, band = 95% bootstrap CI. Final-norm layer left out.

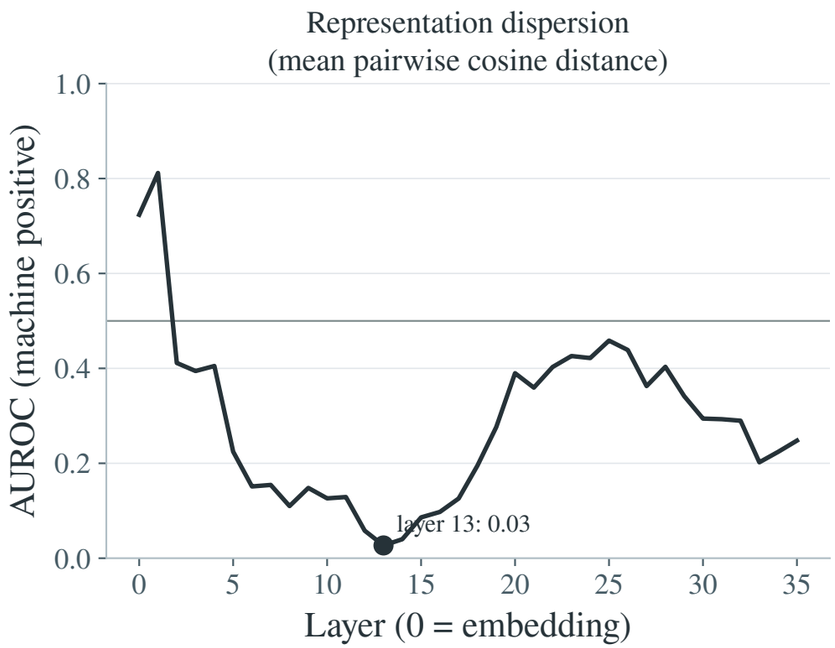
Ministral-8B, drlXDomainLen_wiki_500_tokens: norm relative to the embedding, per text

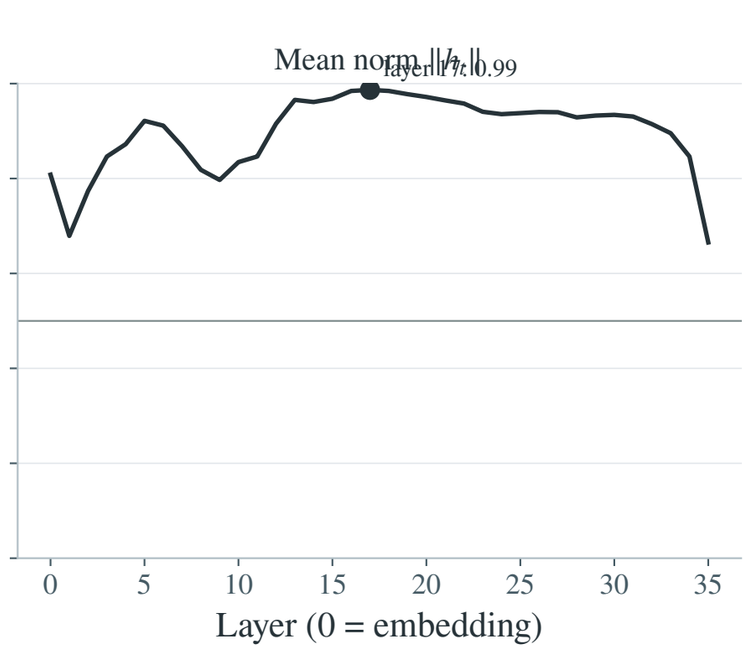
Norm without outlier tokens: per text and layer, tokens with norm > 2 × the text's median token norm are excluded (on average 2.4 tokens per text, 0.46%, layers 1+)

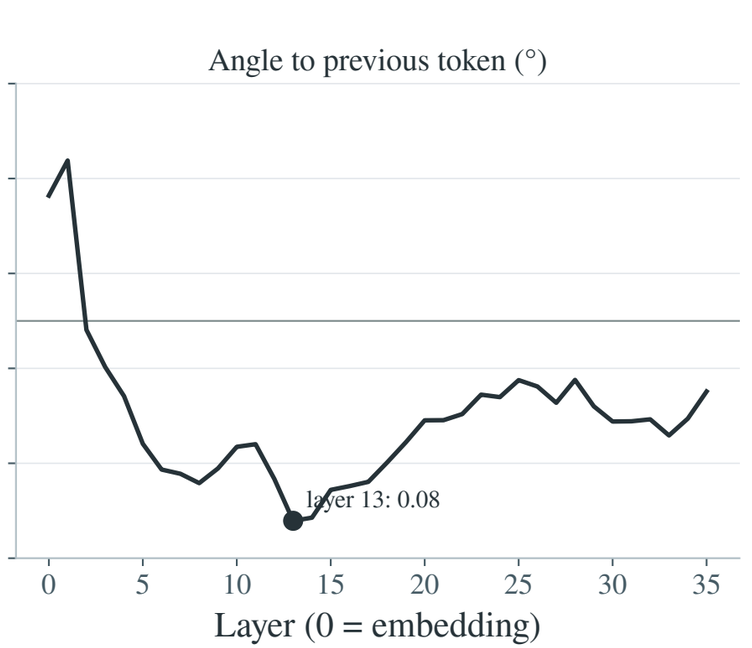


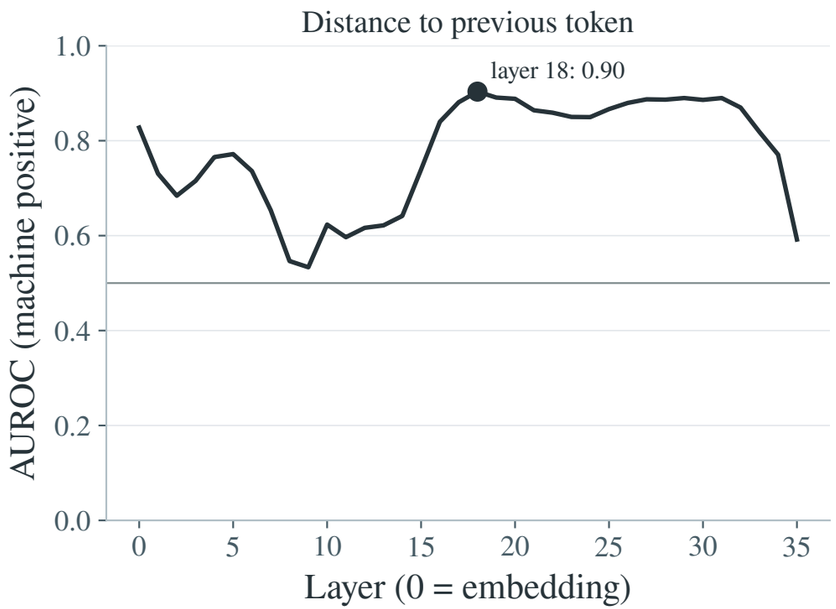
Thin lines = single texts, thick line = group mean. Log y. Final-norm layer left out.

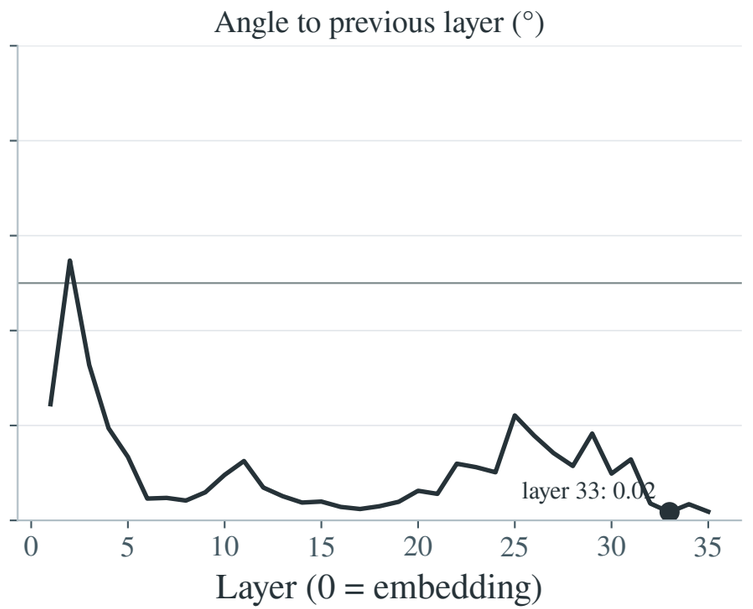
Ministral-8B, drlXDomainLen_wiki_500_tokens: human vs machine separation per layer

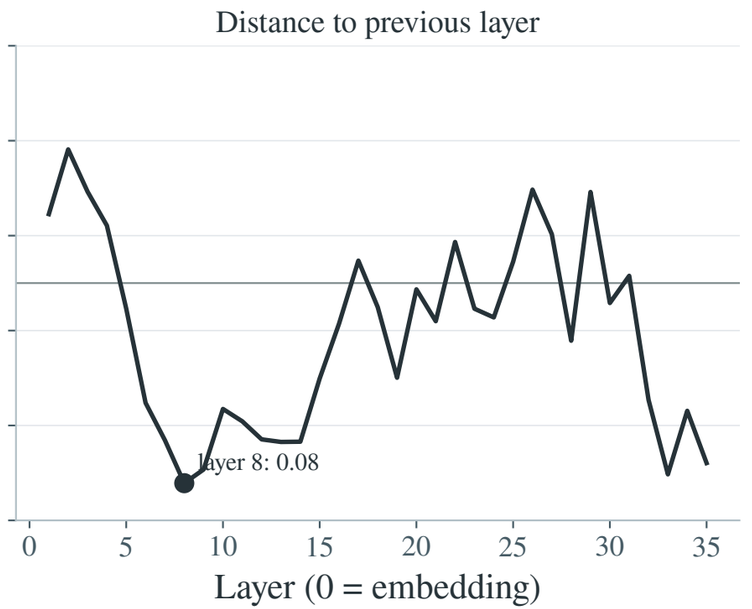




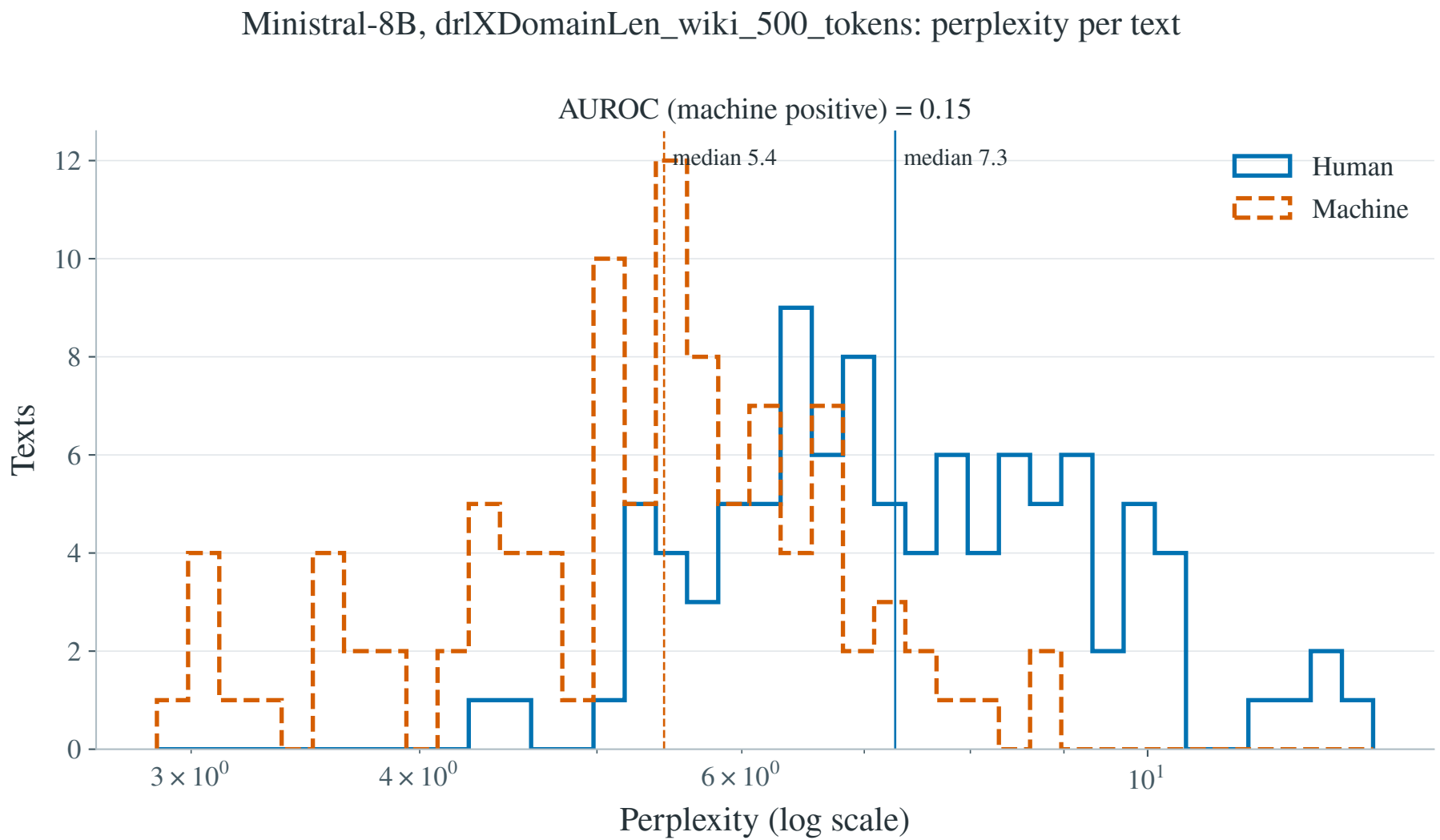




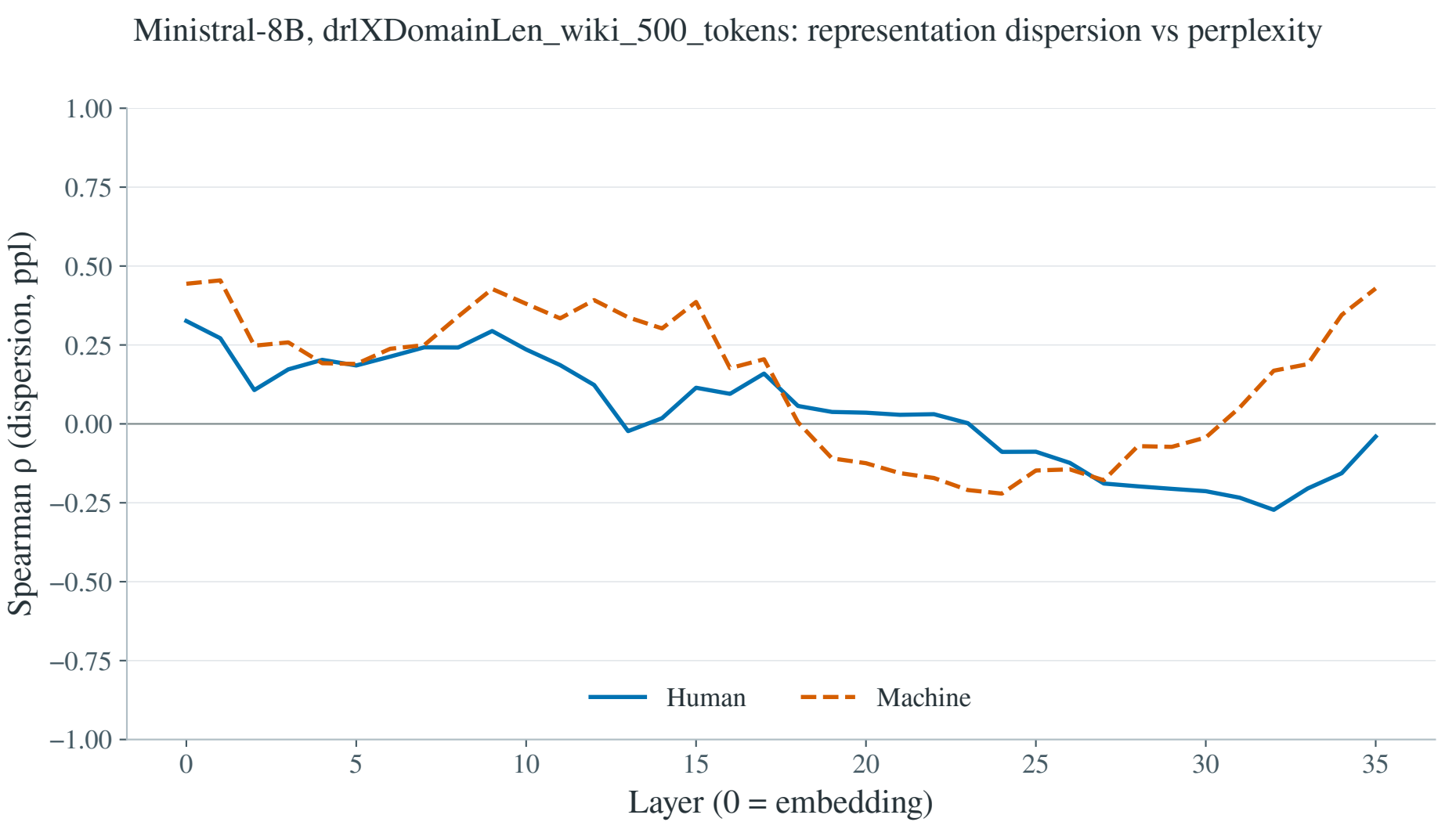




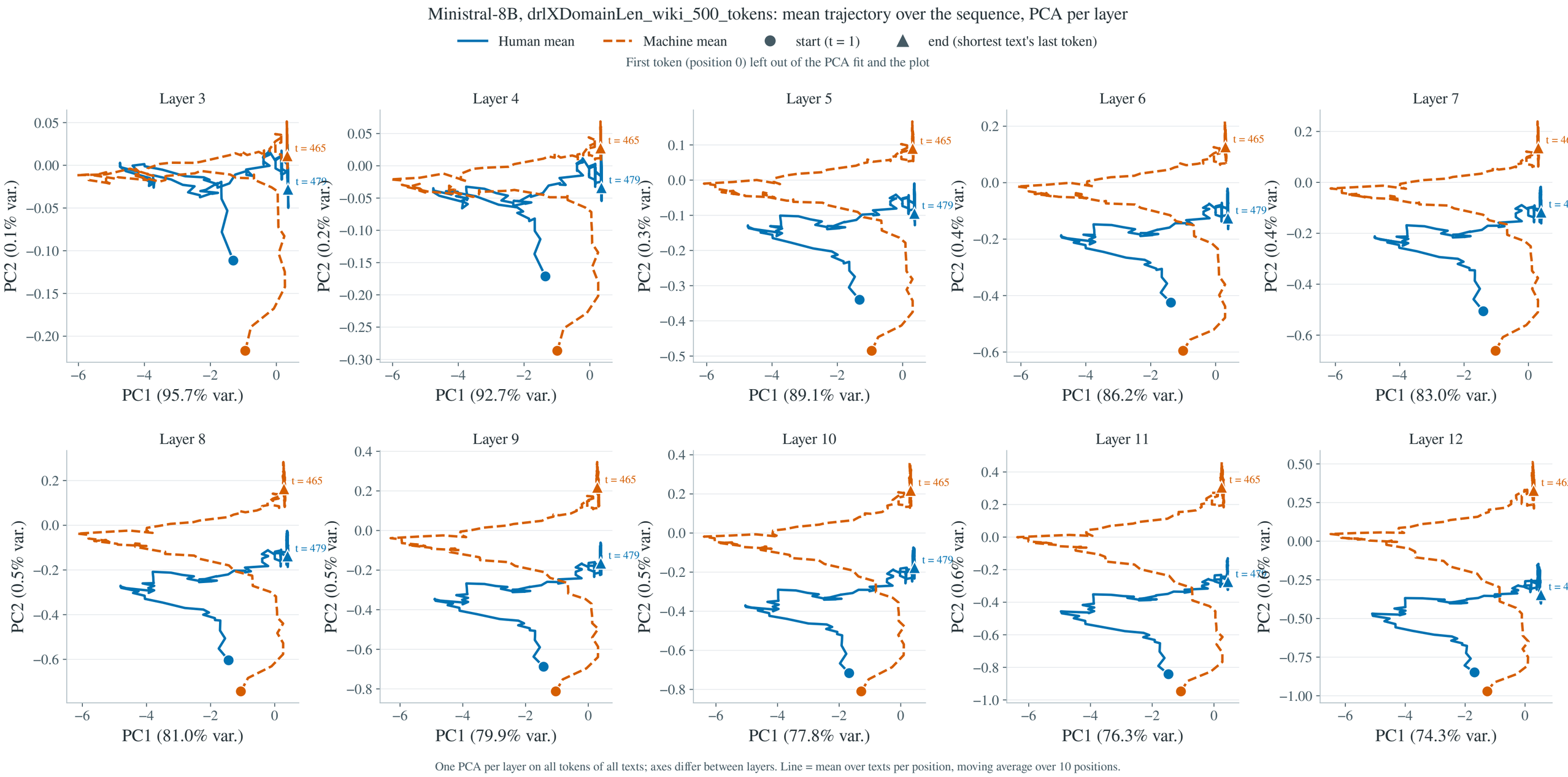
AUROC of each score at each layer; 0.5 = no separation, below 0.5 = lower for machine text. Dot = layer furthest from 0.5.



Histogram on log-spaced bins; vertical lines = medians.



Per layer and label, rank correlation over texts between each text's dispersion and its ppl. Final-norm layer left out.

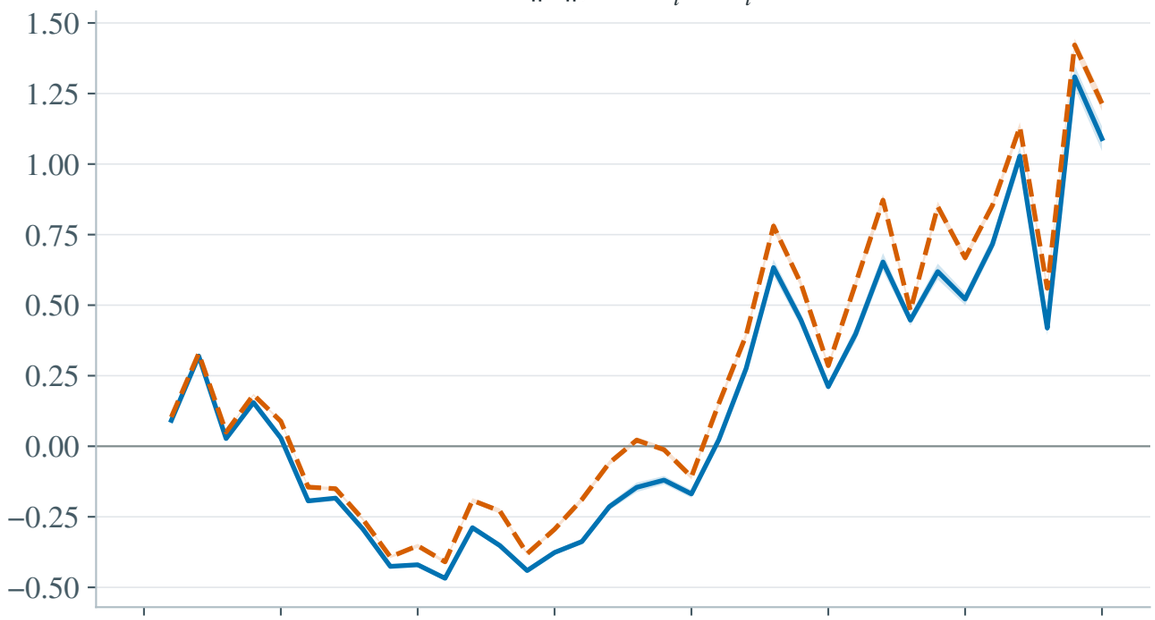

Ministral-8B, drlXDomainLen_wiki_500_tokens: mean trajectory over the sequence, PCA per layer
Human mean
Machine mean
start (t = 1)
end (shortest text's last token)
First token (position 0) left out of the PCA fit and the plot
Layer 3
PC1 (95.7% var.)
PC2 (0.1% var.)
Layer 4
PC1 (92.7% var.)
PC2 (0.2% var.)
Layer 5
PC1 (89.1% var.)
PC2 (0.3% var.)
Layer 6
PC1 (86.2% var.)
PC2 (0.4% var.)
Layer 7
PC1 (83.0% var.)
PC2 (0.4% var.)
Layer 8
PC1 (81.0% var.)
PC2 (0.5% var.)
Layer 9
PC1 (79.9% var.)
PC2 (0.5% var.)
Layer 10
PC1 (77.8% var.)
PC2 (0.5% var.)
Layer 11
PC1 (76.3% var.)
PC2 (0.6% var.)
Layer 12
PC1 (74.3% var.)
PC2 (0.6% var.)
t = 465
t = 479
One PCA per layer on all tokens of all texts; axes differ between layers. Line = mean over texts per position, moving average over 10 positions.

Ministral-8B, drlXDomainLen_wiki_500_tokens: change split along and across the previous state

Human Machine

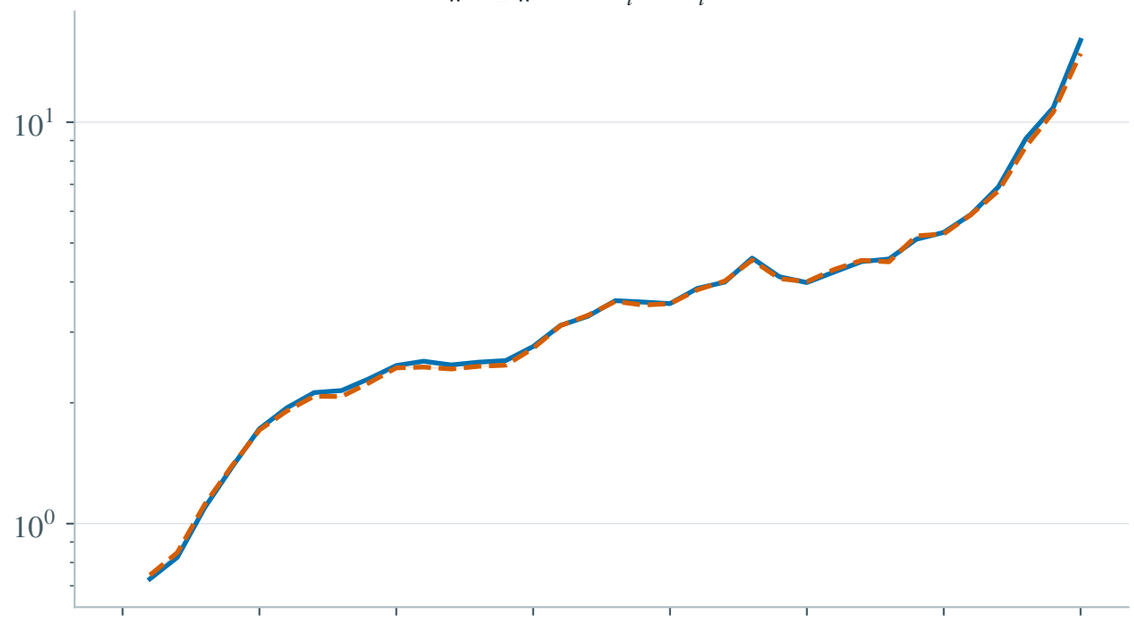
Across layers: parallel

$\Delta \cdot h / \|h\|$, $\Delta = h_t^\ell - h_t^{\ell-1}$



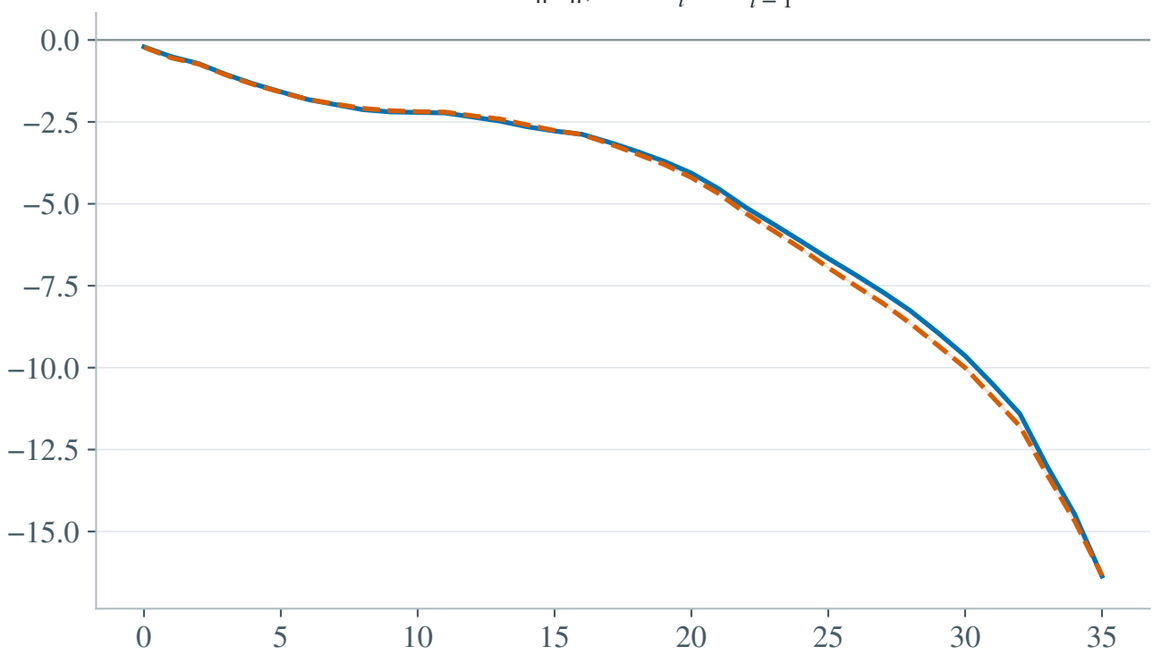
Across layers: perpendicular

$\|\Delta_\perp\|$, $\Delta = h_t^\ell - h_t^{\ell-1}$



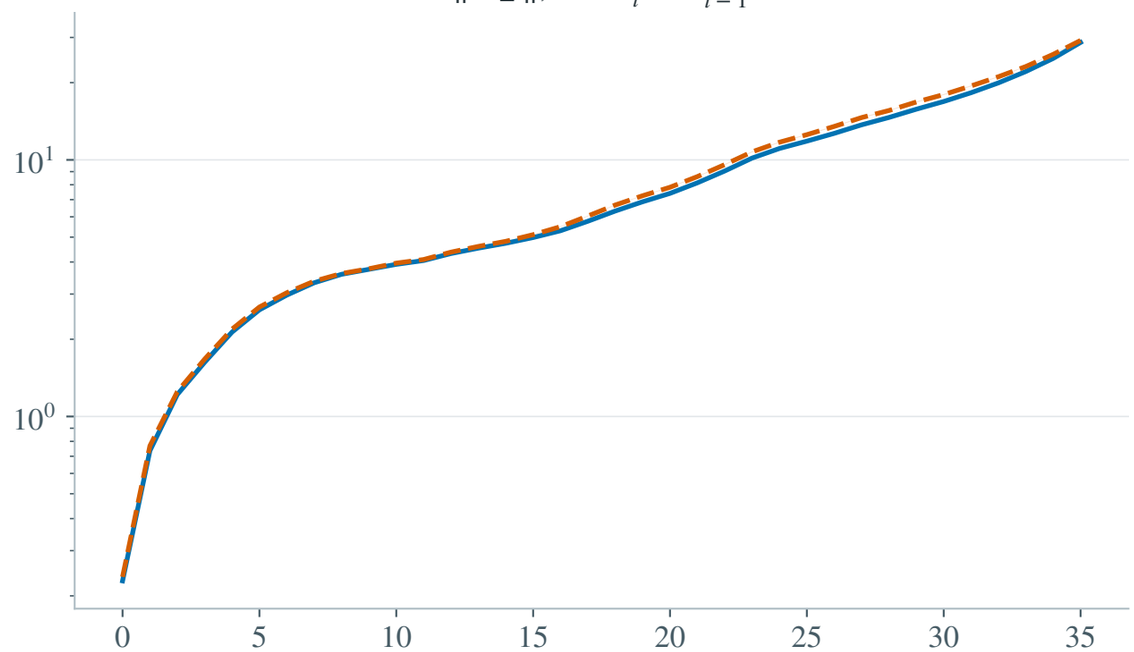
Within a layer: parallel

$\Delta \cdot h / \|h\|$, $\Delta = h_t^\ell - h_{t-1}^\ell$



Within a layer: perpendicular

$\|\Delta_\perp\|$, $\Delta = h_t^\ell - h_{t-1}^\ell$

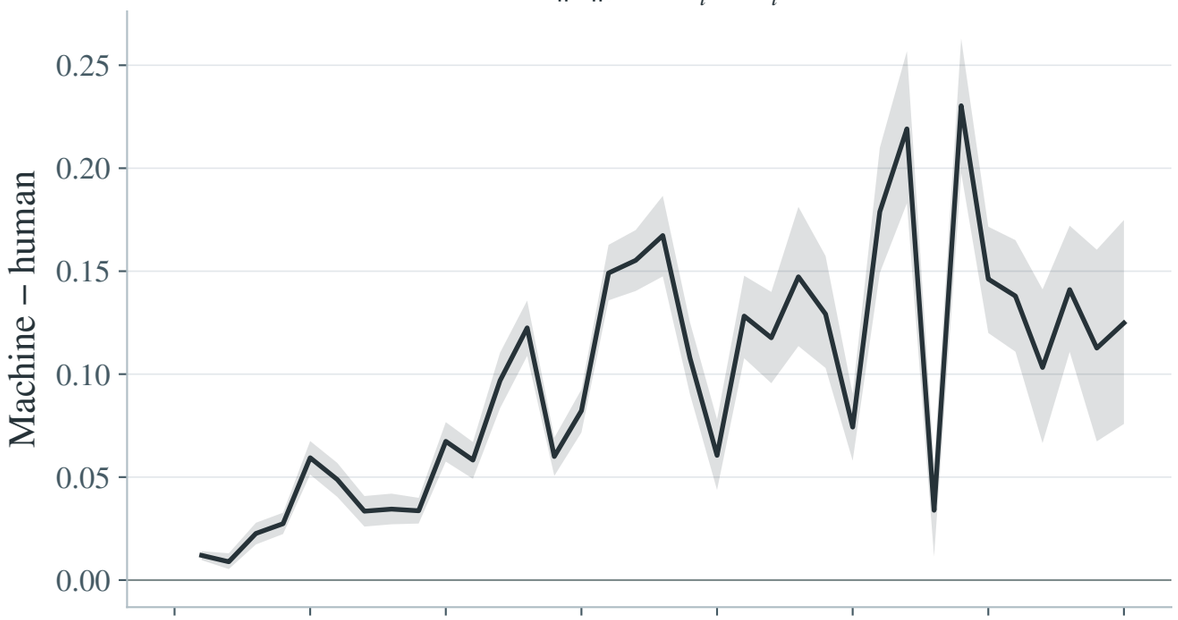


Per text the mean over tokens, then mean over texts, band = 95% CI. h = the previous state; parallel > 0 = grows along h. Tokens with norm > 2 × the text's median norm (at either end of Δ) left out. Log y for perpendicular. Final-norm layer left out.

Ministral-8B, drlXDomainLen_wiki_500_tokens: machine − human gap in the change parts

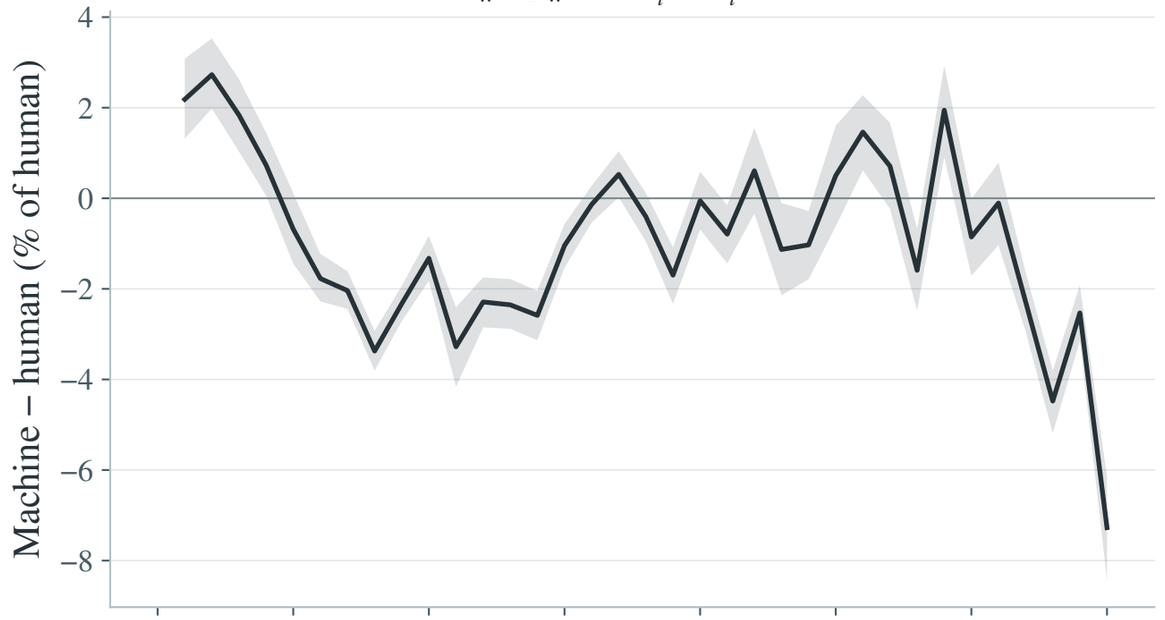
Across layers: parallel

$\Delta \cdot h / \|h\|$, $\Delta = h_t^\ell - h_t^{\ell-1}$



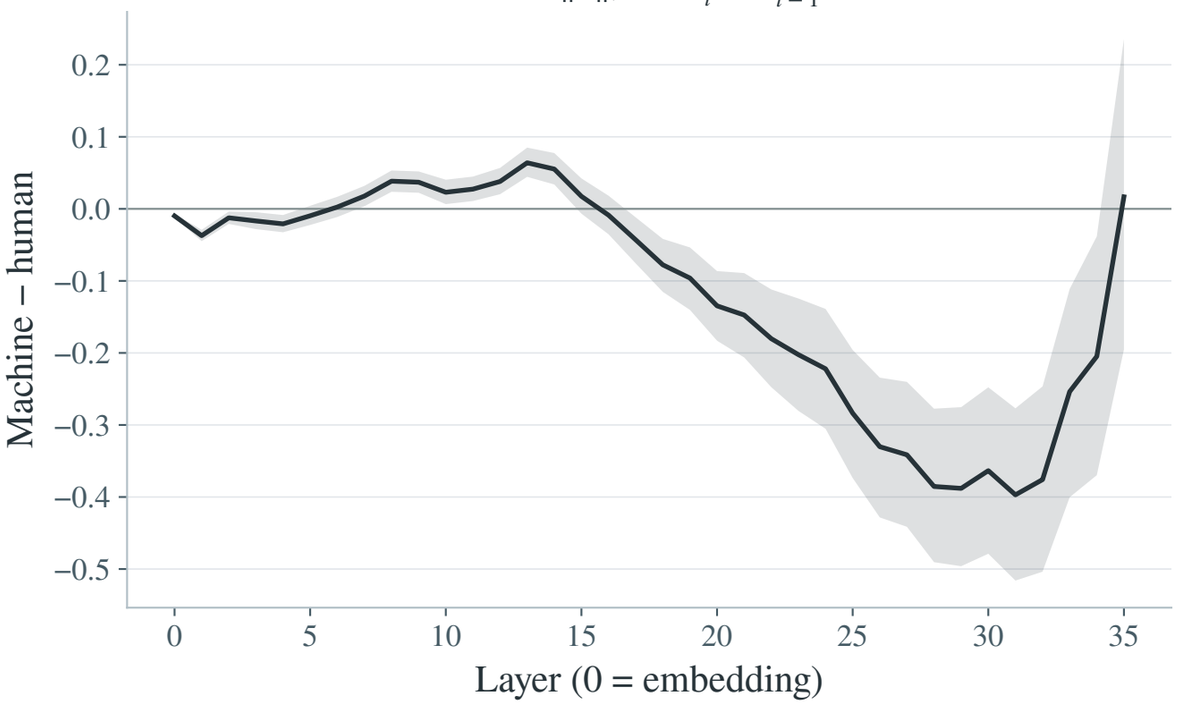
Across layers: perpendicular

$\|\Delta_\perp\|$, $\Delta = h_t^\ell - h_t^{\ell-1}$



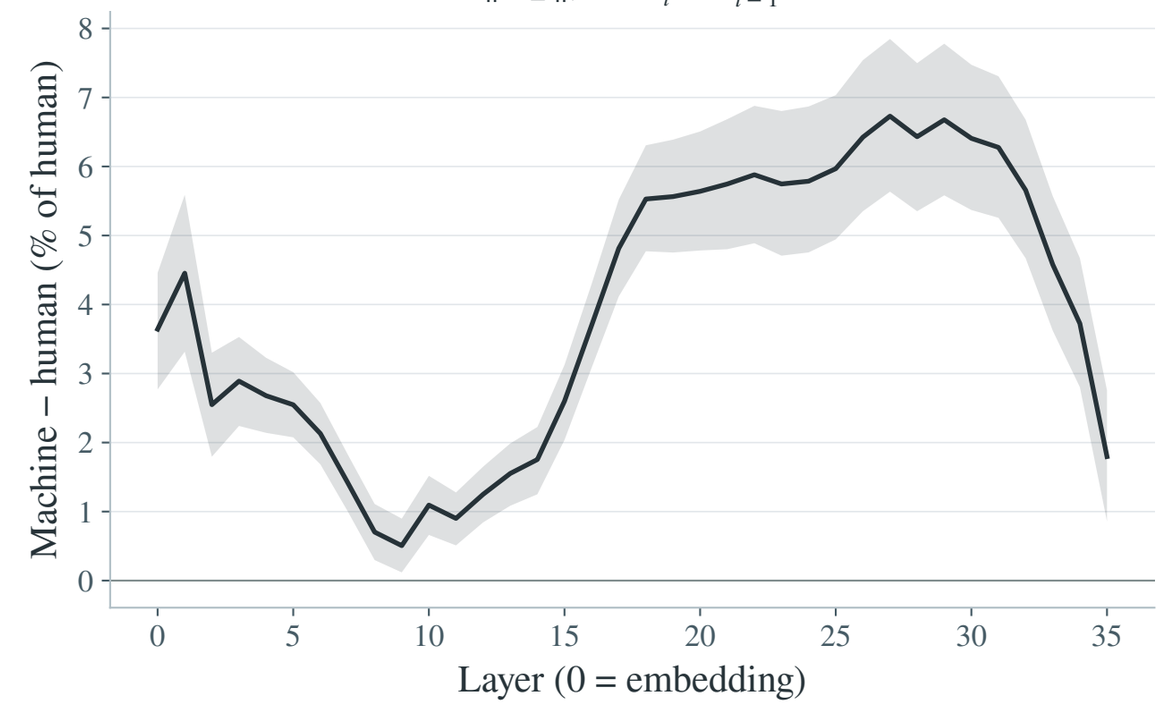
Within a layer: parallel

$\Delta \cdot h / \|h\|$, $\Delta = h_t^\ell - h_{t-1}^\ell$



Within a layer: perpendicular

$\|\Delta_\perp\|$, $\Delta = h_t^\ell - h_{t-1}^\ell$



Gap of the per-text means of page 6, band = 95% bootstrap CI. Parallel: plain difference (it changes sign); perpendicular: % of the human mean. Final-norm layer left out.

Ministral-8B, drlXDomainLen_wiki_500_tokens: human vs machine separation of the change parts

Across layers: parallel

$\Delta \cdot h / \|h\|, \Delta = h_t^\ell - h_t^{\ell-1}$

layer 16: 0.98

Across layers: perpendicular

$\|\Delta_\perp\|, \Delta = h_t^\ell - h_t^{\ell-1}$

layer 8: 0.08

Within a layer: parallel

$\Delta \cdot h / \|h\|, \Delta = h_t^\ell - h_{t-1}^\ell$

layer 1: 0.18

Within a layer: perpendicular

$\|\Delta_\perp\|, \Delta = h_t^\ell - h_{t-1}^\ell$

layer 18: 0.94

AUROC (machine positive)

1.0
0.8
0.6
0.4
0.2
0.0

0 5 10 15 20 25 30 35

Layer (0 = embedding)

AUROC of each per-text mean at each layer; 0.5 = no separation, below 0.5 = lower for machine text. Dot = layer furthest from 0.5.